You will learn how to dynamically scale a fleet of EC2 instances to asynchronously generate screenshots of URLs in the following example. Doing so allows you to guarantee low response times at any time because the load-intensive workload is isolated into background jobs.

Decoupling a dynamic EC2 instance pool asynchronously offers an advantage if you want to scale based on workload: because requests don’t need to be answered immediately, you can put requests into a queue and scale the number of EC2 instances based on the length of the queue. This gives you an accurate metric to scale, and no requests will be lost during a load peak because they’re stored in a queue.

To handle the peak load in the evening, you want to use autoscaling. To do so, you need to decouple the creation of a new bookmark and the process of generating a preview of the website. Chapter 14 introduced an application called URL2PNG that transforms a URL into a PNG image. Figure 17.7 shows the architecture, which consists of an SQS queue for asynchronous decoupling as well as S3 for storing generated images. Creating a bookmark will trigger the following process:

1. A message is sent to an SQS queue containing the URL and the unique ID of the new bookmark.
2. EC2 instances running a Node.js application poll the SQS queue.
3. The Node.js application loads the URL and creates a screenshot.
4. The screenshot is uploaded to an S3 bucket, and the object key is set to the unique ID.
5. Users can download the screenshot directly from S3 using the unique ID.